

Ankit Kumar

Machine Learning Engineer | Agentic AI, RAG and Decision Intelligence

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Seattle, WA | ankitkr8540.github.io/portfolio | U.S. work authorization: F-1 STEM OPT through May 2028

SUMMARY

Machine Learning Engineer who turns ambiguous business needs into reliable systems across GenAI, forecasting, and on-device ML. Clarifies intent, evaluates trade-offs and risks, and aligns engineering with leadership before execution; delivered RAG services with under 2.5-second P95 latency, thermal predictions with 2.2 degrees C maximum RMSE, and data products that cut analysis preparation by 70%.

EXPERIENCE

Milwaukee Electric Tool Corporation

July 2025 - July 2026

Machine Learning Engineer - Front End Innovation

- Served as the sole ML engineer for FEI's first user-facing ML feature, owning development of an 11-parameter, physics-informed three-body thermal prediction system for a chainsaw platform planned for 200K+ units; optimized the model for real-time execution on resource-constrained hardware, achieved a maximum per-profile RMSE of 2.2 degrees C across 30 held-out profiles, and enabled warnings up to 60 seconds before hard shutoff.
- Reduced experiment turnaround from two hours to 40 minutes by building Azure ETL and model-training pipelines that synchronized 1 kHz electrical telemetry with 50 Hz operational and 1 Hz thermal signals, automated preprocessing and model execution, and versioned datasets, parameters, metrics, and artifacts in MLflow.
- Designed a lab-to-field validation framework using time-series replay, load-brake cycle simulation, and held-out battery configurations; evaluated temperature accuracy, threshold-crossing time, and warning repeatability, reducing validation time from 30 days to three.
- Isolated a persistent 4-5 degrees C prediction error to invalid ambient-sensor measurements through feature ablation, counterfactual testing, and telemetry analysis; validated a sensor-independent configuration that supported approximately \$100K in projected annual hardware savings.
- Partnered with the principal firmware engineer to port the Python reference implementation to embedded C for real-time, offline inference without a runtime cloud dependency; built a time-causal reference twin with Streamlit telemetry replay and identified two release-blocking OCV/DCIR and initialization defects before field validation, avoiding an estimated 30+ days of rework.
- Anticipated model-generalization risk across future battery pack configurations and evaluated configuration-aware modeling and additional hardware sensing with the battery team; identified an approximately \$1M cost dependency and presented an alternative adopted by director-level and cross-functional stakeholders as the go-forward strategy.

Mitra | Mason Innovation Exchange, George Mason University

June 2024 - May 2025

Founder and Lead ML Engineer - Selected University Venture

- Translated ambiguous advising needs into an AI-powered university decision-support platform, deploying a Dockerized FastAPI and LangGraph service on ECS Fargate with SageMaker-hosted retrieval and reranking for a controlled alpha; maintained under 2.5-second P95 latency and earned 78% positive ratings from 200+ students.
- Built an event-driven Scrappy, Lambda, SQS, Glue, and PySpark pipeline that processed 8K+ university webpages per crawl, removed approximately 70% of low-value content, and refreshed 50K+ source-traceable chunks in under 30 minutes; implemented policy-aware PII redaction while retaining approved public institutional contact information.
- Designed a bounded LangGraph workflow that decomposed multi-intent questions into subqueries, retrieved evidence for each information need, and routed insufficient evidence to retry, clarification, or abstention; limited requests to five agent actions and two LLM calls to control latency, cost, and failure propagation.
- Increased Precision@10 from 67% to 92% on a fixed 50-query benchmark by combining MiniLM embeddings, FAISS HNSW, lexical search, weighted reciprocal rank fusion, and cross-encoder reranking.
- Established evaluation-driven CI/CD gates using Pytest, GitHub Actions, and DeepEval LLM-as-judge metrics for faithfulness, context relevance, and answer relevance; evaluated toxicity, policy violations, hallucinations, and misinformation before deploying approved images to ECS, with versioned ECR and S3 artifacts supporting rollback.
- Built an offline preference-learning loop from controlled-alpha thumbs-up/down feedback, LangSmith traces, and CloudWatch metrics; trained a DistilBERT reward model and applied trIX PPO to refine response alignment while using feedback-derived hard negatives to retrain the cross-encoder.

Accenture

February 2021 - July 2023

Packaged App Development Analyst

- Built governed PySpark and SQL pipelines on Amazon S3 to standardize point-in-time fundamentals for approximately 9,500 public companies into partitioned Parquet; partnered with quantitative analysts and business stakeholders to reduce company and competitor analysis preparation from five business days to 1.5 days, a 70% reduction.
- Worked with upstream data-engineering owners to correct missing quarters, irregular fiscal periods, and report-date inconsistencies; established point-in-time rules that separated fiscal period from information availability, preserved reported values, and prevented look-ahead leakage in reports and model evaluation.
- Evaluated pooled LSTM, XGBoost, and per-company SARIMA against accuracy, stability, and interpretability requirements; automated two-year KPI forecasts, improving directional accuracy from 64% to 78% in stable periods and from 52% to 61% during transitions.
- Diagnosed a stakeholder-raised gap in structured forecasts and secured business, financial, and product stakeholder alignment for a FinBERT signal trained on approximately 75K timestamped earnings calls, news, and analyst reports; achieved 0.76 macro-F1 on a chronological holdout, raised transition accuracy from 61% to 66%, and published forecasts through an API.

PROJECTS

PyLinguist: Cross-Lingual Python Translation

Research Project | Submitted to ACL BEA

November 2024 - Present

- Built a two-stage Python translation system across seven languages using token-aware translation, a curated dictionary of 234 Python terms, and GPT-4o or Claude 3.5 Haiku refinement while preserving code syntax and structure.
- Evaluated 70 held-out programs using BLEU, syntax validity, structural preservation, semantic similarity, and bilingual review; GPT-4o achieved a 92.5% composite score versus 61.5% for the translation-only baseline, with 100% syntax validity.

TECHNICAL SKILLS

Languages and Data: Python, SQL, C, PySpark, Pandas, NumPy, ETL, Parquet

GenAI and Agents: RAG, LangGraph, LangChain, question decomposition, tool orchestration, prompt engineering, evidence-gated generation, retry, clarification, abstention, RLHF

Retrieval and NLP: SentenceTransformers, FAISS HNSW, hybrid and lexical search, weighted RRF, cross-encoder reranking, FinBERT, DistilBERT, Hugging Face Transformers

Evaluation and Responsible AI: DeepEval, LLM-as-judge, online and offline evaluation, faithfulness, context relevance, answer relevance, Precision@K, Recall@K, Hit Rate@K, NDCG@K, hallucination, misinformation, toxicity, policy validation, PII redaction

Machine Learning: PyTorch, scikit-learn, XGBoost, LSTM, SARIMA, time-series forecasting, physics-informed modeling

Cloud and MLOps: AWS, SageMaker, Lambda, S3, SQS, Glue, Athena, ECR, ECS Fargate, CloudWatch, Azure, MLflow, LangSmith

Deployment and Engineering: Edge ML, on-device inference, embedded C, FastAPI, Docker, GitHub Actions, Pytest, CI/CD, REST APIs, Streamlit, Pydantic

EDUCATION

George Mason University

M.S. in Computer Science - Machine Learning Concentration | GPA: 3.90/4.00

Distinguished Academic Achievement Award, College of Engineering and Computing

August 2023 - May 2025

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate (MLA-C01) | AWS Certified Developer - Associate (DVA-C02)